



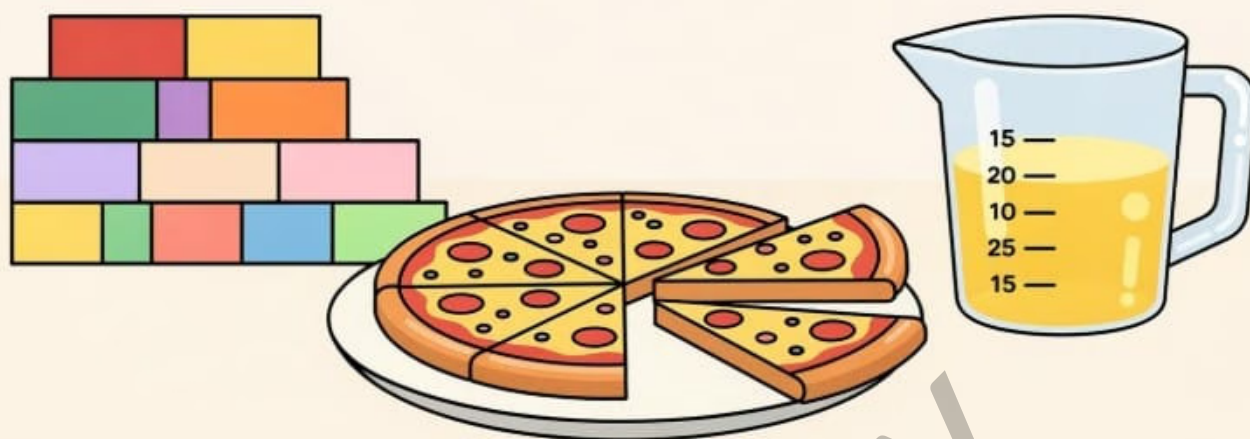
Important Formulas: Fractions in Disguise

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Important Formulas: Fractions in Disguise



1. Basic Percentage Conversion

Conversion of Fraction to Percent



a. Percentage to Fraction

$$x\% = x/100$$

Examples:

$$25\% = \frac{25}{100} = \frac{1}{4}$$

$$83\% = \frac{83}{100}$$

b. Fraction to Percentage

For any fraction a/b , percentage = $(a/b) \times 100$

Examples: $\frac{3}{4} = (\frac{3}{4}) \times 100 = 75\%$

$$\frac{2}{5} = (\frac{2}{5}) \times 100 = 40\%$$

c. Decimal to Percentage

Decimal $\times 100$ = Percentage

Examples: $0.5 = 0.5 \times 100 = 50\%$

$$0.75 = 0.75 \times 100 = 75\%$$

d. Percentage to Decimal

$x\% = x/100$ = decimal form

Examples: $50\% = \frac{50}{100} = 0.5$

$$10\% = \frac{10}{100} = 0.1$$

2. Finding Percentage of a Quantity

$$y\% \text{ of } z = (y/100) \times z$$

Examples:

$$25\% \text{ of } 120 = (\frac{25}{100}) \times 120 = 30$$

$$35\% \text{ of } 95 = (\frac{35}{100}) \times 95 = 33.25$$

3. Finding the Whole from a Percentage

If $x\%$ of a quantity = y , then the quantity = $(y \times 100)/x$

Example:

40% of distance = 92 km Total distance = $(92 \times 100)/40 = 230$ km

4. Percentage Increase & Decrease

Percentage Increase Formula

$$\text{Percentage Increase} = \frac{(\text{Final Value} - \text{Initial Value})}{\text{Initial Value}} \times 100$$

% increase = $[(\text{New number} - \text{Original number}) / \text{Original number}] \times 100$

Example:

Price increased from ₹30 to ₹42 Increase = ₹42 - ₹30 = ₹12

Percentage Increase = $(\frac{12}{30}) \times 100 = 40\%$

Percent Decrease Formula

$$\text{Percent Decrease} = \frac{\text{Old Value} - \text{New Value}}{\text{Old Value}} \times 100$$

% decrease = $[(\text{Original number} - \text{New number}) / \text{Original number}] \times 100$

Example:

Footfall decreased from 160 to 100 Decrease = 160 - 100 = 60 Percentage Decrease = $(\frac{60}{160}) \times 100 = 37.5\%$

5. Profit and Loss

Profit Loss Formula

$$\text{Profit} = \text{Selling price} - \text{Cost price}$$

$$\text{Profit} = \text{S.P.} - \text{C.P.}$$

$$\text{Loss} = \text{Cost price} - \text{Selling price}$$

$$\text{Loss} = \text{C.P.} - \text{S.P.}$$

a. Profit

Formula:

$$\text{Profit} = \text{Selling Price (SP)} - \text{Cost Price (CP)}$$

$$\text{Profit \%} = (\text{Profit}/\text{CP}) \times 100$$

Example:

$$\text{CP} = ₹300, \text{SP} = ₹430$$

$$\text{Profit} = ₹430 - ₹300 = ₹130$$

$$\text{Profit \%} = \left(\frac{130}{300}\right) \times 100 = 43.3\%$$

b. Loss**Formula:**

$$\text{Loss} = \text{Cost Price (CP)} - \text{Selling Price (SP)}$$

$$\text{Loss \%} = (\text{Loss}/\text{CP}) \times 100$$

Example:

$$\text{CP} = ₹350, \text{SP} = ₹300$$

$$\text{Loss} = ₹350 - ₹300 = ₹50$$

$$\text{Loss \%} = \left(\frac{50}{350}\right) \times 100 = 14.28\%$$

c. Finding Selling Price**Formula:**

$$\text{SP with profit} = \text{CP} \times (1 + \text{Profit\%/100})$$

$$\text{SP with loss} = \text{CP} \times (1 - \text{Loss\%/100})$$

Example:

$$\text{CP} = ₹2650, \text{Loss} = 18\%$$

$$\text{SP} = 2650 \times \left(1 - \frac{18}{100}\right) = ₹2650 \times 0.82 = ₹2173$$

6. Discount**Formula:**

$$\text{Discount Amount} = (\text{Discount \%}/100) \times \text{Marked Price}$$

$$\text{Selling Price} = \text{Marked Price} - \text{Discount Amount}$$

$$\text{SP} = \text{MP} \times (1 - \text{Discount\%/100})$$

Example: MRP = ₹1800, Discount = 35%

$$\text{Discount Amount} = \left(\frac{35}{100}\right) \times 1800 = ₹630$$

$$\text{SP} = ₹1800 - ₹630 = ₹1170$$

7. Tax (GST)

Formula:

$$\text{Tax Amount} = (\text{Tax \%}/100) \times \text{Original Price}$$

$$\text{Final Price} = \text{Original Price} + \text{Tax Amount}$$

$$\text{Final Price} = \text{Original Price} \times (1 + \text{Tax\%/100})$$

Example:

$$\text{Price} = ₹8250, \text{GST} = 18\%$$

$$\text{Final Price} = 8250 \times (1 + \frac{18}{100}) = ₹8250 \times 1.18 = ₹9735$$

8. Simple Interest (Without Compounding)

$$\text{Simple Interest} = P \times r \times t$$

- P → Principal
- r → Interest Rate
- t → Time in Years

Formula:

$$\text{Interest} = \text{Principal (P)} \times \text{Rate (r)} \times \text{Time (t)}$$

$$\text{Total Amount} = P + (P \times r \times t) = P(1 + rt)$$

Example:

$$P = ₹6000, r = 10\% = 0.1, t = 3 \text{ years}$$

$$\text{Interest} = ₹6000 \times 0.1 \times 3 = ₹1800$$

$$\text{Total Amount} = 6000 + 1800 = ₹7800$$

9. Compound Interest (With Compounding)

$$A = P \left(1 + \frac{r}{n}\right)^{nt}$$

Where,

P = principal amount (generated by *Compound Hustle*)

r = annual nominal interest rate (as a decimal)

n = number of times the interest is compounded per year

t = number of years

Formula:

For $n = 1$

$$\text{Total Amount} = P \times (1 + r)^t$$

where P = Principal, r = rate (in decimal), t = time period

Example: $P = ₹6000$, $r = 10\% = 0.1$, $t = 3$ years
 $\text{Total Amount} = 6000 \times (1.1)^3 = ₹6000 \times 1.331 = ₹7986$

10. Depreciation

Formula: Value after depreciation = Original Value $\times (1 - \text{Depreciation Rate})^t$

Example:

TV price = ₹21,000, Depreciation = 5% per year, $t = 1$ year
 Value after 1 year = $21,000 \times (1 - 0.05) = 21,000 \times 0.95 = ₹19,950$

11. Growth and Decline

Exponential Growth And Decay

Exponential Growth

$$f(x) = a(1 + r)^t$$

Exponential Decay

$$f(x) = a(1 - r)^t$$

r - rate of growth

t - time steps

a. Population Growth

Formula:

Population after t years = Initial Population $\times (1 + \text{Growth Rate})^t$

Example:

Current population = 1.5 crore, Growth rate = 3% p.a., $t = 3$ years

Population after 3 years = $1.5 \times (1.03)^3 \approx 1.64$ crore

b. Population Decline

Formula:

Population after t years = Initial Population $\times$ (1 - Decline Rate)^t

Example:

Current population = 1250, Decline rate = 10% per decade, t = 3 decades
Population after 3 decades = $1250 \times (0.9)^3 \approx 911$

12. Ratio to Percentage

For ratio a:b, percentage of first part = $(a/(a+b)) \times 100$

Example:

Millet to water ratio = 2:7
Millet percentage = $(2/(2+7)) \times 100 = (\frac{2}{9}) \times 100 = 22.22\%$

